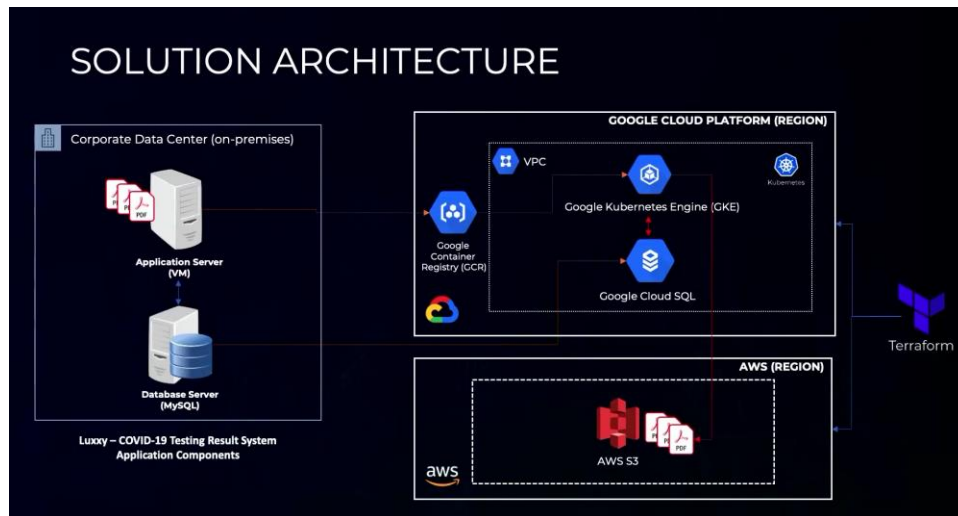


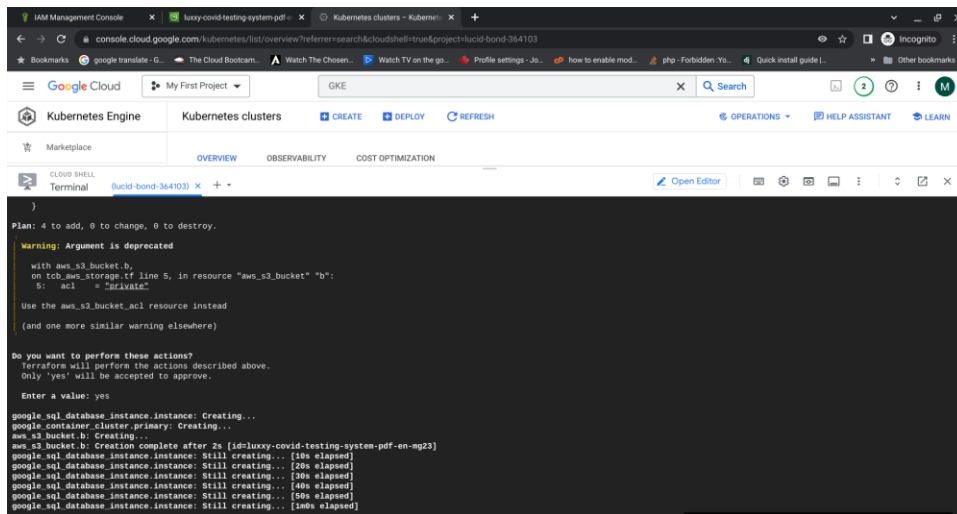
Project Evidence-

Project: Migration of Luxxy - Luxury Hotels & Resorts Application and its Data from On-premises environment to a multi-cloud structure environment (GCP and AWS).

❖ SOLUTION ARCHITECTURE

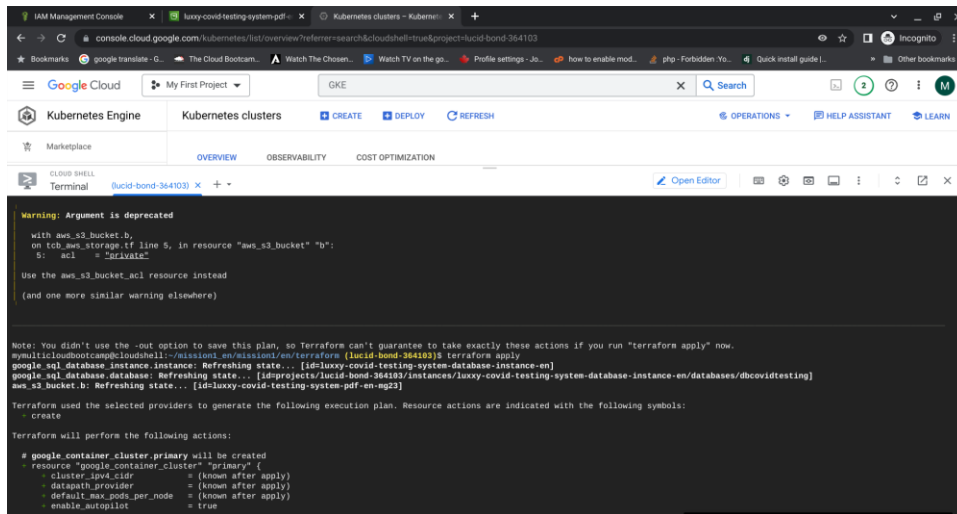


- I. **MISSION 1:** Enable a **MultiCloud architecture** deployment through **Terraform**, with resources running in **AWS** and **Google Cloud Platform**
 - a) Running terraform commands to provision AWS S3 bucket and Google Cloud SQL database Instance and Database and Google Kubernetes Cluster



The screenshot shows the Google Cloud console interface with the 'Kubernetes Engine' section selected. The 'Kubernetes clusters' tab is active, displaying a list of clusters. Below the console, a terminal window shows the output of a Terraform apply command. The output includes a warning about a deprecated argument, a confirmation prompt, and a list of resources being created or updated. The resources listed are:

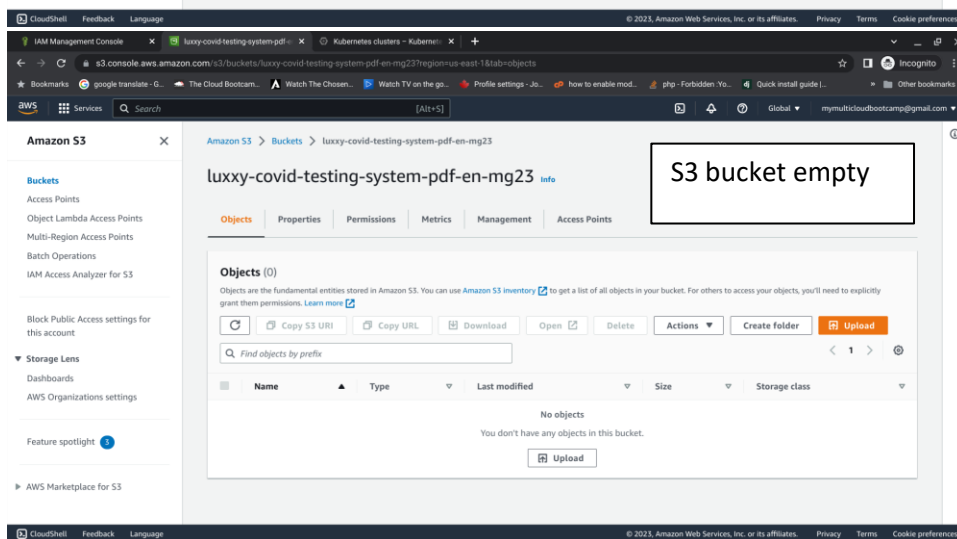
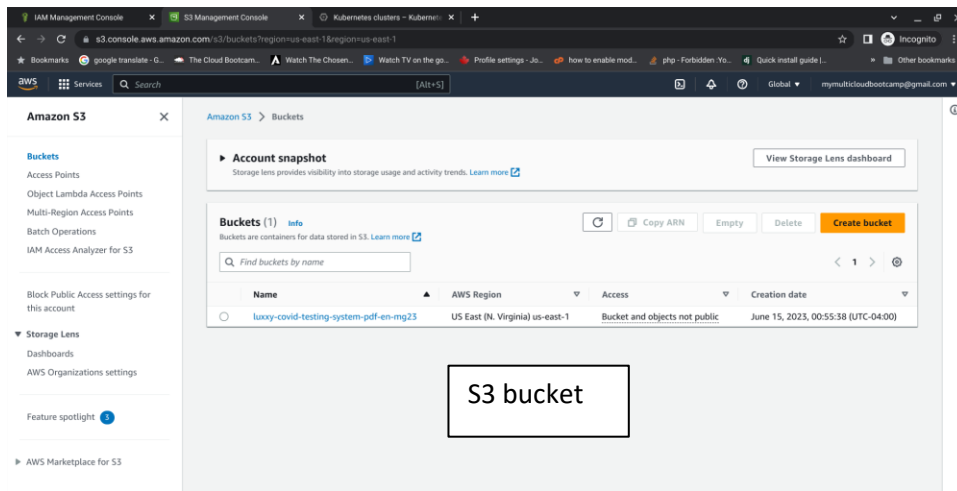
- google.sql_database_instance: Creating...
- google.container_cluster.primary: Creating...
- aws.s3_bucket.b: Creating...
- aws.s3_bucket.b: Creation complete after 2s [id:luxxy-covid-testing-system-pdf-en-mg23]
- google.sql_database_instance: Still creating... [18s elapsed]
- google.sql_database_instance: Still creating... [20s elapsed]
- google.sql_database_instance: Still creating... [38s elapsed]
- google.sql_database_instance: Still creating... [48s elapsed]
- google.sql_database_instance: Still creating... [56s elapsed]
- google.sql_database_instance: Still creating... [18s elapsed]



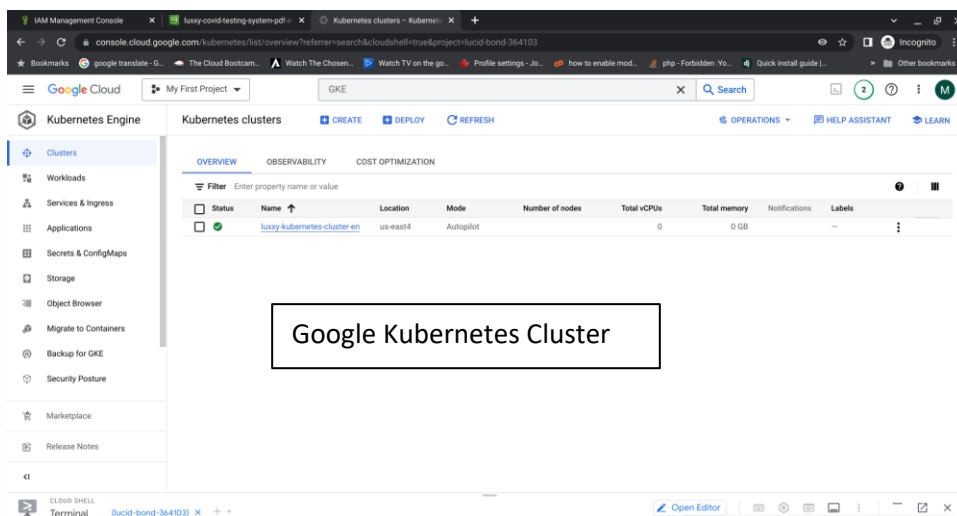
The screenshot shows the Google Cloud console interface with the 'Kubernetes Engine' section selected. The 'Kubernetes clusters' tab is active, displaying a list of clusters. Below the console, a terminal window shows the output of a Terraform apply command. The output includes a warning about a deprecated argument, a note about the .out option, and a list of resources being created or updated. The resources listed are:

- google.sql_database_instance: Refreshing state... [id:luxxy-covid-testing-system-database-instance-en]
- google.sql_database_instance: Refreshing state... [id:projects/lucid-bond-364103/instances/luxxy-covid-testing-system-database-instance-en/databases/dbcovidtesting]
- aws.s3_bucket.b: Refreshing state... [id:luxxy-covid-testing-system-pdf-en-mg23]

b) S3 bucket is provisioned and empty



c) Kubernetes Engine/Kubernetes Cluster is provisioned



d) Google Cloud SQL Database instance and Database are provisioned and connection is configured

Instance ID	Type	Public IP address	Private IP address	Instance connection name	High availability	Location	Storage used	Actions
luxy-covid-testing-system-database-instance-en	MySQL 5.7	34.85.251.66	10.57.96.3	luxid-bond-364103-u...	ENABLE	us-east4-a	1 GB of 10 Gi	

Google Cloud SQL Database Instance

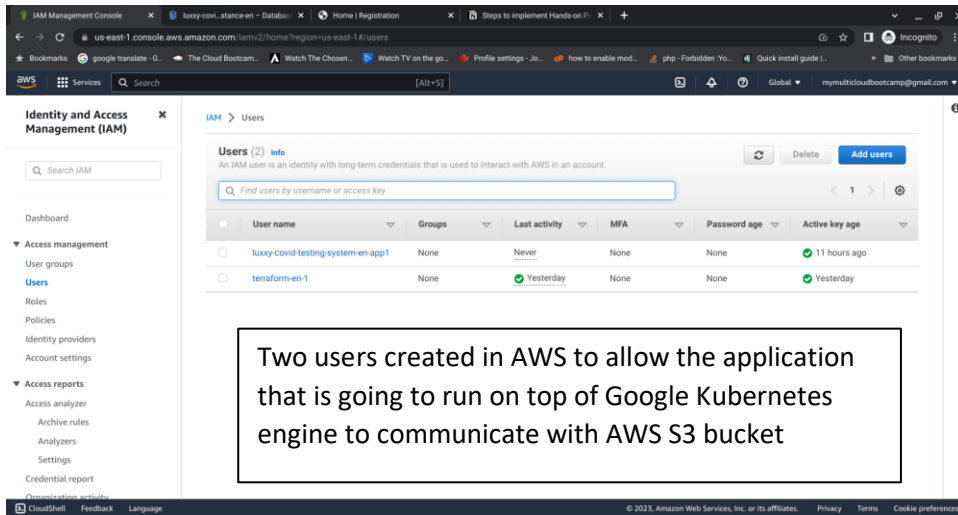
Name	Collation	Character set	Type
dbcovidtesting	utf8_general_ci	utf8	User
information_schema	utf8_general_ci	utf8	System
mysql	utf8_general_ci	utf8	System
performance_schema	utf8_general_ci	utf8	System
sys	utf8_general_ci	utf8	System

Google Cloud SQL Database created

Google Cloud SQL connection is configured.

- II. **MISSION 2:** Process to convert a database and an application to run on the **MultiCloud architecture (AWS and Google Cloud)**, with **Docker** and **Kubernetes**.

a) Creating a user in AWS

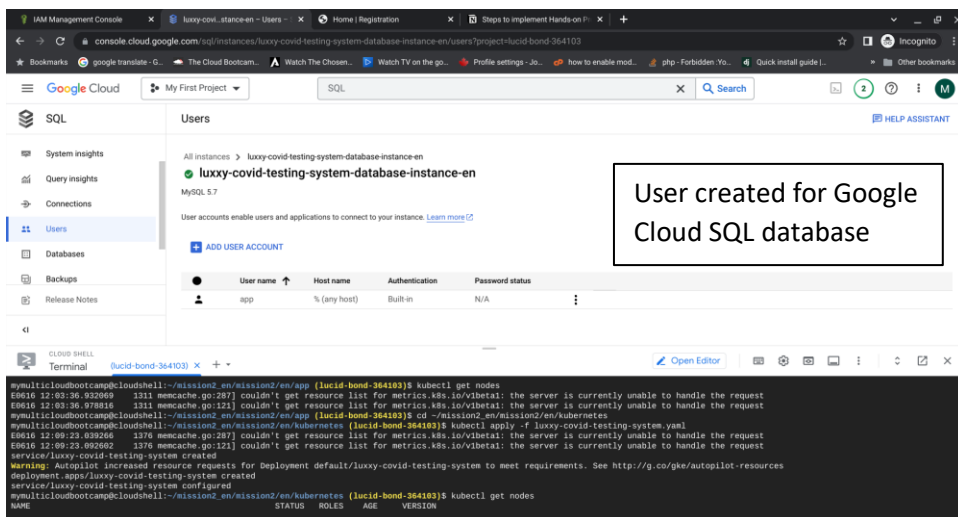


The screenshot shows the AWS IAM Management Console. On the left is a navigation menu with options like Dashboard, Access management, and Access reports. The main area is titled 'IAM > Users' and shows a list of two users:

User name	Groups	Last activity	MFA	Password age	Active key age
luxy-covid-testing-system-en-app1	None	Never	None	None	11 hours ago
terraform-en-1	None	Yesterday	None	None	Yesterday

A text box overlay on the screenshot states: "Two users created in AWS to allow the application that is going to run on top of Google Kubernetes engine to communicate with AWS S3 bucket".

b) Creating a Google Cloud SQL database user



The screenshot shows the Google Cloud SQL console. The left sidebar has a 'Users' section selected. The main area shows details for a MySQL instance named 'luxy-covid-testing-system-database-instance-en'. Below this, there is a table of users:

User name	Host name	Authentication	Password status
app	% (any host)	Built-in	N/A

A text box overlay on the screenshot states: "User created for Google Cloud SQL database".

Below the table, there is a 'CLOUD SHELL' terminal window showing commands and output for setting up the database and application.

c) Running terraform and MySQL commands to create a Google Cloud SQL Database table and to deploy an application on top of Google Kubernetes Engine/cluster.

Application deployed. View from the kubernetes cluster

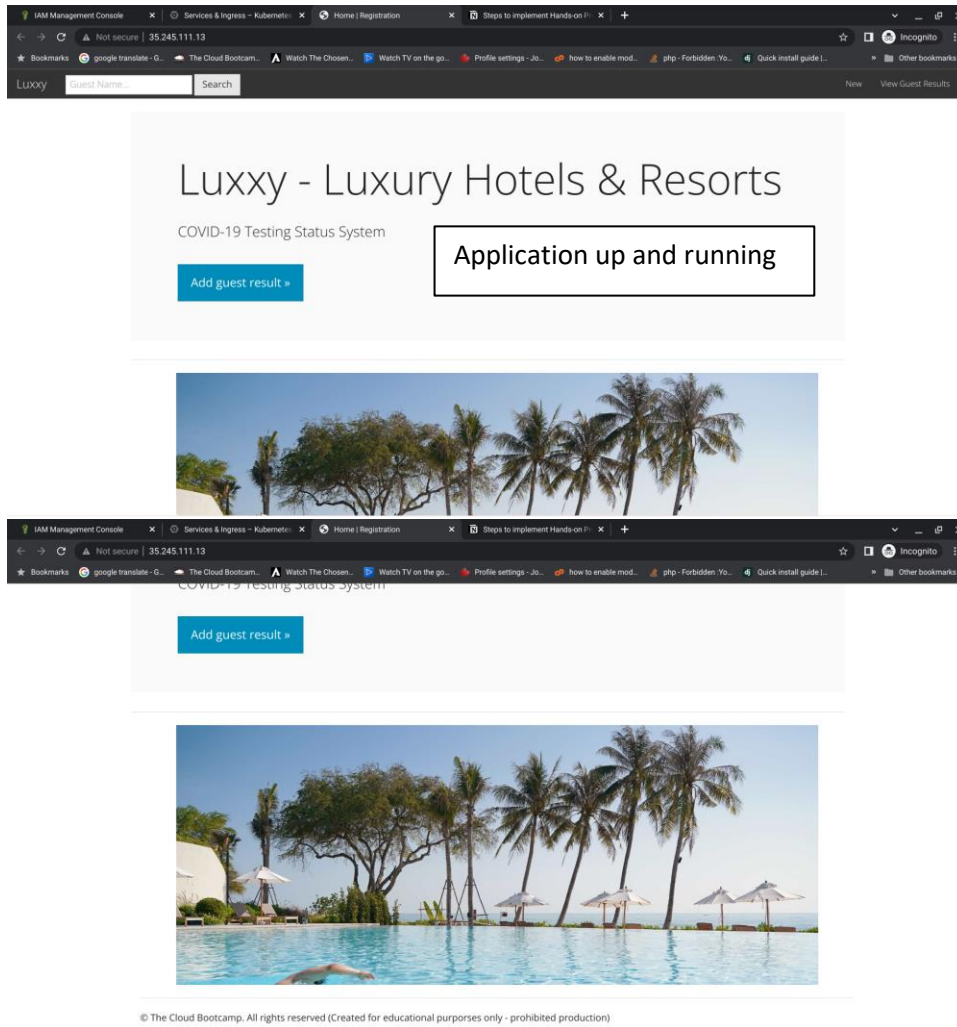
View from the kubernetes workloads

View from the kubernetes services and Ingress

Terminal output (repeated across all screenshots):

```
E0816 12:09:23.039266 1378 memcache.go:287] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
E0816 12:09:23.092602 1378 memcache.go:121] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
Warning: Autopilot increased resource requests for Deployment default/luxy-covid-testing-system to meet requirements. See http://g.co/gke/autopilot-resources
deployment.apps/luxy-covid-testing-system created
service/luxy-covid-testing-system configured
mymlcloudbootcamp@cloudshell:~$ kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
gke3-luxy-kubernetes-cluster-e-pool-1-87b9b0dd-slvq  Ready    <none>   4m54s  v1.26.8-gke.500
mymlcloudbootcamp@cloudshell:~$ kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
luxy-covid-testing-system-7f7bdfb8-rxg5  1/1      Running   0           6m5s
mymlcloudbootcamp@cloudshell:~$ kubectl get pods -n default
NAME                                READY    STATUS    RESTARTS   AGE
luxy-covid-testing-system-7f7bdfb8-rxg5  1/1      Running   0           6m5s
```

d) The application is up and running on top of a Google Kubernetes Engine/Cluster



III. **MISSION 3: Professionally migrate the application files and data** from on-premises application server and database server

a) Migration of database data to Google Cloud SQL database

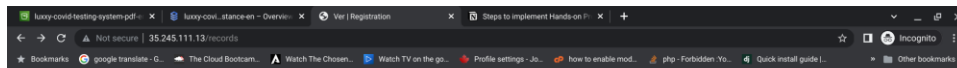
Database data migrated to Google Cloud SQL database

b) Migration of PDF files to AWS S3 bucket

PDF files migrated to AWS S3 bucket

c) The Application and its data have been successfully migrated from on-premises environment to a multi-cloud structure environment (AWS and GCP). The **application** is up and running on top

of Google Kubernetes Engine and all the data has been migrated to AWS S3 bucket and Google Cloud SQL database



The Luxxy - Luxury Hotels & Resorts application up and running with all its data.

Records

Guest Name	Home Country	Testing Type	Testing Result	Scanned Result (PDF)	
Amanda Colema	USA	Antigen	Negative	Link	Edit Delete
Ann Sharp	USA	Antigen	Negative	Link	Edit Delete
Annie Meyer	USA	Antigen	Negative	Link	Edit Delete
Beverly Cohen	USA	Antigen	Negative	Link	Edit Delete
Billie Balidai	USA	Antigen	Negative	Link	Edit Delete
Brandi Mathis	USA	Antigen	Negative	Link	Edit Delete
Charlene Brya	USA	Antigen	Negative	Link	Edit Delete
Clifford Gray	USA	Antigen	Negative	Link	Edit Delete
Dallas Maxwell	USA	Antigen	Negative	Link	Edit Delete
Darryl Gregor	USA	Antigen	Negative	Link	Edit Delete

Services & Ingress - Kubernetes - covid-testing.pdf

luxxy-covid-testing-system-pdf-en-mg23.s3.amazonaws.com/v029s02-as02020c09s-s29312-as-amanda.pdf?AWSAccessKeyId=AKIA555GHHMUYH66FQJMS&Signature=NuYTY2F64HXAO3Kjd...

1 / 1 | 100% +

Viewing one of the migrated the PDF files.

SUMMIT MEDICAL GROUP
1 Diamond Hill Road
Berkeley Heights, NJ 07922
LABORATORY DEPARTMENT

31D0116144
Gordana Katava, DO

Name/DOB: TESTPATIENT, MARGUERITE (6/14/1995)
Patient ID: EH13
Phone number: (908) 278-4344
Procedure Date: 6/10/2020 4:23 PM
Approval date: 6/10/2020 4:26 PM

Sex: F Fasting:
Age: 24
Home Chart Location:
Encounter ID:

Provider: Provider Test
Draw Location: BH_1Diamond_LAB
Case Number: 2016202006
Technician: Kahal Patel
Entered by: Kahal Patel

TEST NAME	RESULT	UNITS	REFERENCE RANGE
	IN RANGE	OUT OF RANGE	
COVID-19 Antigen	NEGATIVE		KP

COVID-19 Antigen
06/10/20 4:25 PM

A negative test result does not rule out the possibility of an infection with SARS-CoV-2. Serum or plasma samples from the early (pre-seroconversion) phase of illness can yield negative findings. Therefore, this test cannot be used to diagnose an acute infection. Testing with molecular diagnostic should be performed to evaluate for active infection in symptomatic individuals. It is not known at this time if the presence of antibodies to SARS-CoV-2 confers immunity to reinfection. False positive results for the test may occur due to cross-reactivity. The Siemens Covid-19 Total Antibody test was developed, and its performance characteristics determined by Siemens for the Vista 1500. The FDA has authorized this test for EUA.